

When Similarity Misleads: A Diagnostic Benchmark for Chinese Text Rewriting Evaluation

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Abstract

Chinese text rewriting evaluation is difficult because a good rewrite must preserve meaning while changing expression, making source-rewrite similarity an unreliable proxy for quality. We introduce **RewritingBench**, a diagnostic benchmark of 730 Chinese source-rewrite pairs with human quality ratings from three trained annotators ($\text{ICC}(2,1) = 0.87$). Using this benchmark, we show that seven commonly used source-similarity metrics, including lexical overlap and embedding-based similarities, correlate negatively with human scores. This reveals a systematic mismatch: similarity metrics reward staying close to the source, while humans reward successful transformation under meaning preservation. We then train RewriteJudge, a task-specific evaluator for Chinese rewrite quality. Task-specific training substantially improves alignment with human rankings over zero-shot and general-purpose judges. Cross-source pairwise aggregation achieves the highest ranking correlation in our benchmark ($\rho = 0.688$).

1 Introduction

A rewrite should be semantically faithful but expressively different. We focus on *general-purpose Chinese text improvement rewriting*, where the goal is to improve clarity, fluency, and readability while preserving the source meaning. This setting differs from reference matching tasks because high-quality rewrites are expected to depart from the source wording, so surface overlap is not a reasonable proxy for quality.

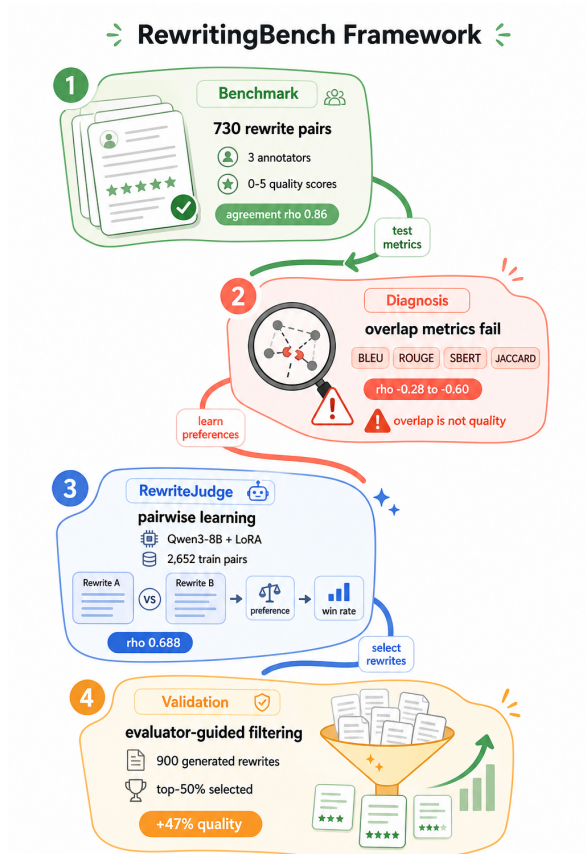


Figure 1: Overview of RewritingBench: benchmark construction, diagnostic analysis of the mismatch between similarity metrics and human quality scores, cross-source pairwise preference evaluation, and preliminary model-scored filtering analysis.

Standard metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) rely on surface overlap, while embedding similarities such as SBERT COSINE reward semantic closeness. For rewriting, however, strong outputs often need to move away from the wording of the source. This raises a concrete research question: *when the desired output should differ from the input, are source-*

rewrite similarity metrics measuring quality or merely measuring similarity? The question is especially open for Chinese rewriting, where we are not aware of a human-rated diagnostic benchmark specifically targeting general-purpose rewrite quality despite growing interest in Chinese LLM evaluation (Huang et al., 2023).

We organize this work around three research questions. **RQ1:** Can trained annotators reliably judge general-purpose Chinese rewrite quality? **RQ2:** Do source-rewrite similarity metrics align with human rewrite quality judgments? **RQ3:** Does task-specific evaluator training, especially pairwise preference aggregation, improve global ranking of rewrite quality compared with direct score prediction and zero-shot judging?

To answer these questions, we present **RewritingBench** (Figure 1), a diagnostic benchmark and analysis of Chinese rewriting evaluation. Our contributions are:

1. We introduce RewritingBench, a human-rated diagnostic benchmark for Chinese general-purpose rewriting evaluation.
2. We show that source-rewrite similarity metrics are negatively aligned with human rewrite quality, revealing a systematic mismatch between similarity and transformation quality.
3. We train RewriteJudge and show that task-specific supervision improves ranking alignment; cross-source pairwise aggregation achieves the highest ranking correlation ($\rho = 0.688$).

2 Related Work

2.1 Source Similarity and Reference-Based Metrics

Traditional rewriting evaluation relies on overlap metrics like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), while embedding metrics such as BERTScore (Zhang et al., 2020), MoverScore (Zhao et al., 2019), BARTScore (Yuan et al., 2021), and SBERT (Reimers and Gurevych, 2019) improve semantic matching through contextualized representations. Methods such as ParaScore (Lan and Chen, 2023) and Alam et al. (2024) consider paraphrase quality

and multiple dimensions, and prior surveys document broader obstacles in NLG evaluation (Gehrmann et al., 2023; Gao et al., 2025). These methods mostly treat similarity as a useful proxy for quality. We do not claim to exhaust all automatic evaluation methods; instead, we focus on a diagnostic question: whether source-rewrite similarity, a common proxy in rewriting evaluation, aligns with human quality judgments. Our work tests this assumption in Chinese rewriting, where desirable transformation may make similarity and quality conflict.

2.2 LLM-as-a-Judge and Task-Specific Evaluators

The LLM-as-a-Judge paradigm offers a flexible alternative to reference metrics. G-Eval (Liu et al., 2024) uses prompting for structured evaluation, Prometheus 2 (Kim et al., 2024) fine-tunes a 7B model for absolute and pairwise scoring, and Chatbot Arena (Chiang et al., 2024) leverages human pairwise comparisons. However, general-purpose judges can exhibit bias and rating inconsistency (Wang et al., 2024a,b; Lee et al., 2025). We examine whether such judges transfer to the specialized, subjective task of Chinese rewrite quality evaluation and whether task-specific training is necessary.

2.3 Pairwise Preference and Ranking-Based Evaluation

Pairwise comparison is often more stable than absolute scoring because it asks for a relative judgment rather than global calibration. Kim et al. (2024) find higher agreement with human judgments in general NLP tasks, Zheng et al. (2023) use pairwise comparison for LLM assessment, and work on generative selection shows that comparative scoring can improve selection from N candidates (Wang et al., 2024c). Preference learning work further shows that models can learn from pairwise signals without explicit reward scores (Rafailov et al., 2023; Lambert et al., 2024). We test whether this comparative signal specifically helps Chinese rewriting in a cross-source global ranking setting, where human quality judgments are subjective and ordinal.

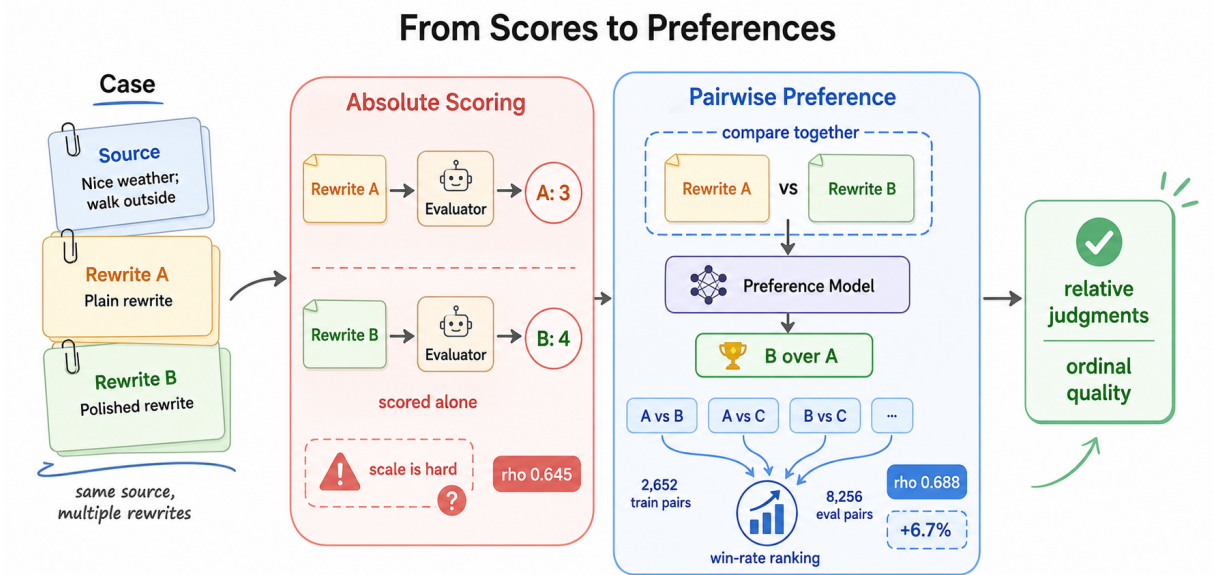


Figure 2: Conceptual illustration of absolute scoring and pairwise preference evaluation. Absolute scoring judges each rewrite independently on a fixed scale, whereas pairwise preference evaluation compares candidate rewrites jointly and aggregates comparison outcomes into a win rate ranking. The pairwise setting in our benchmark compares source-rewrite pairs across different sources for global quality ranking.

3 Benchmark and Evaluation Protocol

Task definition. Given a Chinese source text s and a candidate rewrite r , the evaluator should judge whether r improves expression while preserving the source meaning. A high-quality rewrite should be faithful, fluent, natural, and appropriately transformed. Transformation is valuable only when it improves clarity, readability, or expression; unnecessary changes that distort meaning or reduce naturalness should be penalized. Our benchmark focuses on *general-purpose text improvement rewrites* that enhance fluency, clarity, or readability without targeting a specific style. Formally, the absolute scoring formulation is:

$$f : \mathcal{S} \times \mathcal{R} \rightarrow [0, 5] \quad (1)$$

3.1 Evaluation Framework

We define text rewriting quality along four complementary dimensions, scored from 0 to 5: **Meaning preservation** (preserve the source meaning without factual errors or omissions of key content), **Expression improvement** (improve fluency, clarity, and readability), **Appropriate transformation** (make necessary and effective structural, phrasing, and syntactic adjustments, rather than mechanical sub-

stitution or excessive rewriting), and **Naturalness and idiomacity** (ensure the Chinese expression is natural and contextually appropriate). Since RewritingBench focuses on general-purpose rewriting rather than a predefined target style, we use expression improvement instead of style fidelity as a core criterion. The final score is a holistic judgment integrating all dimensions. In our protocol, annotators provide only this unified rating, using the four dimensions as internal guidelines rather than separate scores. We do not reward divergence for its own sake: a rewrite that is lexically or structurally different but semantically distorted or less natural should receive a low score.

3.2 Benchmark Construction

We collected 730 Chinese rewrite pairs, each consisting of one source text and one generated rewrite. Each source text appears once in the benchmark (730 unique inputs), and the training and evaluation split has no exact source overlap. Source texts contain 49 to 251 Chinese characters (mean 103.2, median 93), while rewrites contain 20 to 247 characters (mean 93.4, median 84.5). Three trained annotators independently scored each pair on the holistic scale

from 0 to 5. Agreement is high across multiple statistics: mean pairwise Spearman $\rho = 0.88$, ICC(2,1)=0.87, Krippendorff’s alpha (ordinal)=0.87, mean weighted kappa (quadratic)=0.87, exact agreement=81.1%, and ± 1 agreement=94.1%. We use the average rating as the gold standard. The dataset is split into 600 training pairs and 130 evaluation pairs (129 after removing one parsing issue) using stratified sampling by score. The evaluation set closely reflects the full score distribution: 0 (15.5%), 1 (27.9%), 2 (24.8%), 3 (20.2%), 4 (9.3%), 5 (2.3%). Because the current version does not contain complete source-domain and generator metadata, we treat RewritingBench as a diagnostic benchmark rather than a comprehensive coverage benchmark.

3.3 Class Balanced Training Strategy

The score distribution is imbalanced (Score 0: 15%, Score 5: 3%). When we train RewriteJudge on the original data, 82% of its predictions collapse to Score 1 and the correlation with human scores is low ($\rho = 0.253$). We therefore oversample minority score classes and create a balanced training set of 1,008 samples (168 per class). After balancing, the model predicts across the full score range and better matches human rankings.

3.4 Evaluator Training

We fine-tune Qwen3 8B (Yang et al., 2025) and Qwen2.5 7B Instruct (Team, 2025) with LoRA (Hu et al., 2022). For absolute scoring, the model outputs a single holistic score from 0 to 5. For pairwise preference learning, the model chooses which of two rewrites is higher quality, and comparison outcomes are aggregated into win rate rankings. Full training details and hyperparameters are provided in Appendix A.

4 Diagnostic Analysis

4.1 Evaluation Setup

4.1.1 Evaluation Metrics

All correlations are computed on a held out set of 129 samples to prevent data leakage. We evaluate performance against human gold standards using **Spearman** ρ as the primary ranking metric, **Pearson** r , and **Kendall** τ .

We also report **MAE** / **RMSE**, and **Exact** / ± 1 / ± 2 agreement percentages to measure absolute error and prediction tolerance.

4.1.2 Baselines

We evaluate 19 methods across four categories: **traditional metrics** (BLEU, ROUGE L, SBERT COSINE, TFIDF COSINE, W2V COSINE, JACCARD WORD, JACCARD CHAR), **zero-shot LLMs** (Qwen3 and Qwen2.5, 7B to 14B), **general-purpose evaluators** (G-Eval, Prometheus 2), and **fine-tuned variants** (LoRA with different data sizes and formats). For pairwise evaluation, we additionally test eight zero-shot models from 72B to endpoint-scale, including Qwen2.5 72B Instruct, Qwen3 235B A22B, DeepSeek V3 series, GPT 4.1, and Kimi K2.

4.2 RQ1/RQ2: Alignment with Human Judgment

Table 1 summarizes the benchmark evaluation. For RQ1, the high annotator agreement reported in Section 3 establishes that trained humans can reliably judge Chinese rewrite quality. For RQ2, the results show a sharp divide: task-specific fine-tuning can recover a meaningful signal (RewriteJudge: $\rho = 0.645$), while source-similarity metrics and general-purpose LLM judges do not match the same human scores.

Source-similarity metrics are negatively aligned with human rewrite quality. As shown in Figure 3, all seven source-similarity metrics show a negative correlation with human judgment, from $\rho = -0.28$ (SBERT COSINE) to $\rho = -0.60$ (JACCARD CHAR), with bootstrap 95% confidence intervals that do not cross zero. This occurs because quality rewrites often diverge from the source wording, which overlap metrics penalize. These metrics are not designed to evaluate transformation quality: the negative correlations show that using source similarity as a proxy for rewrite quality can systematically rank copy-like rewrites above successful rewrites. Even SBERT COSINE ($\rho = -0.28$) fails because semantic similarity alone does not distinguish between faithful transformation and mere copying.

Why the failure is systematic. The negative correlations are not random metric

Table 1: Main evaluation results on the Chinese rewriting quality benchmark. Source-similarity metrics are negatively correlated with human scores. Best results are in **bold**.

Method	Type	Spearman ρ	Pearson r	Kendall τ	MAE	RMSE	Exact / ± 1 / ± 2 (%)
RewriteJudge (Ours, Qwen3 8B)	Fine tuned (8B)	+0.645	+0.602	+0.530	0.91	1.24	25.6 / 79.8 / 95.4
+Reasoning prefix	Fine tuned (8B)	+0.360	+0.374	+0.312	1.07	1.36	14.0 / 72.1 / 93.0
Unbalanced training (Qwen3 8B)	Fine tuned (8B)	+0.337	+0.344	+0.289	1.12	1.43	14.0 / 70.5 / 90.7
RewriteJudge (Ours, Qwen2.5 7B)	Fine tuned (7B)	+0.580	+0.580	+0.470	0.94	1.25	24.8 / 75.2 / 93.0
Unbalanced training (Qwen2.5 7B)	Fine tuned (7B)	+0.287	+0.246	+0.238	1.10	1.40	14.0 / 72.1 / 91.5
Prometheus 2	Fine tuned (7B)	+0.145	+0.126	+0.120	1.54	1.95	17.0 / 53.8 / 75.5
Zero shot (Qwen3 14B)	Zero shot (14B)	+0.129	+0.180	+0.112	2.03	2.37	10.9 / 35.7 / 56.6
Zero shot (Qwen2.5 14B)	Zero shot (14B)	+0.058	+0.105	+0.046	2.14	2.49	11.6 / 30.2 / 56.6
Zero shot (Qwen3 8B)	Zero shot (8B)	+0.106	+0.176	+0.093	1.86	2.23	15.5 / 40.3 / 62.8
Zero shot (Qwen2.5 7B)	Zero shot (7B)	-0.019	+0.079	-0.016	2.04	2.38	10.1 / 37.2 / 56.6
G-Eval (Qwen3 8B)	Zero shot (8B)	+0.041	+0.089	+0.033	1.85	2.22	16.3 / 41.1 / 58.1
G-Eval (Qwen2.5 7B)	Zero shot (7B)	-0.145	-0.064	-0.121	2.22	2.62	12.4 / 33.3 / 52.7
SBERT COSINE	Traditional	-0.280	-0.072	-0.199	NA	NA	NA
BLEU	Traditional	-0.294	-0.211	-0.243	NA	NA	NA
W2V COSINE	Traditional	-0.336	-0.099	-0.238	NA	NA	NA
ROUGE L	Traditional	-0.385	-0.396	-0.317	NA	NA	NA
JACCARD WORD	Traditional	-0.538	-0.509	-0.416	NA	NA	NA
TFIDF COSINE	Traditional	-0.571	-0.503	-0.433	NA	NA	NA
JACCARD CHAR	Traditional	-0.595	-0.550	-0.453	NA	NA	NA

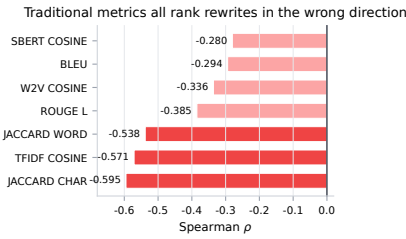


Figure 3: Source-similarity metrics are negatively aligned with human rewrite quality. All seven metrics show negative Spearman ρ with bootstrap 95% CIs that do not cross zero.

noise; they follow from two recurring error patterns (detailed examples in Appendix G). *Surface similarity without quality*: rewrites that are close copies receive high overlap scores because they preserve source wording, but humans assign low scores because little rewriting occurred. *Quality without surface similarity*: strong rewrites that restructure syntax and vary vocabulary receive low overlap scores, even when they preserve meaning and improve fluency. Thus, overlap metrics optimize source similarity, whereas human judgment rewards transformation that preserves meaning. For rewriting, these objectives can be inversely correlated.

General-purpose LLM evaluators underperform in this setup. As shown in Figure 4, we tested Prometheus 2 (Kim et al., 2024) ($\rho = 0.145$) and G-Eval (Liu et al., 2024) ($\rho = 0.041$), as well as zero-shot Qwen3 14B ($\rho = 0.129$). All are below our fine-tuned 8B model. This shows that general-purpose eval-

uator tuning does not transfer reliably to Chinese rewrite quality assessment without task-specific training. This result does not mean that RewriteJudge is a general-purpose rewriting evaluator; it shows that task-specific supervision is necessary for this benchmark and task definition.

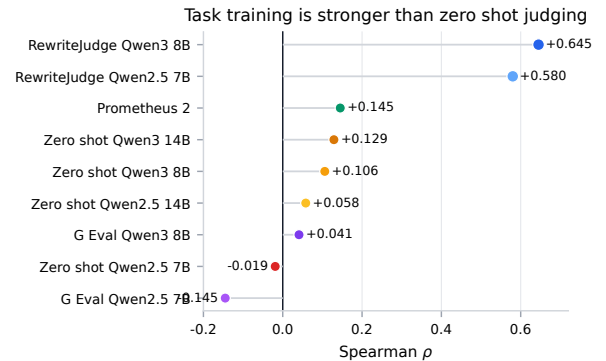


Figure 4: Spearman ρ for LLM-based evaluators. Task-specific RewriteJudge training yields higher correlation than the tested general-purpose and zero-shot evaluators.

5 Cross-Source Pairwise Ranking for Rewrite Quality

RQ3 asks which learning signal best matches human rewrite quality judgments. Absolute scoring requires the model to place each candidate on a global scale from 0 to 5, whereas pairwise preference evaluation asks a simpler comparative question: which of two rewrites is better? Figure 2 illustrates how the same rewriting case is processed under these two

paradigms. In our current benchmark, each source has one rewrite, so pairwise preference is constructed across different source texts rather than among multiple rewrites of the same source.

5.1 Pairwise Construction and Aggregation

We create a global quality ranking set from the 600 training samples. For each pair, the two rewrites must differ by at least 0.67 points in human score:

$$\mathcal{P} = \{(r_i, r_j) \mid |q_i - q_j| \geq 0.67\} \quad (2)$$

yielding 2,652 balanced comparisons (1,326 per class). Because the paired rewrites come from different source texts, each training example presents both the source and rewrite for candidate A and the source and rewrite for candidate B. The model then chooses which candidate demonstrates higher overall rewrite quality relative to its own source.

At inference, all $\binom{129}{2} = 8,256$ evaluation pairs are scored; each rewrite’s *win rate* is:

$$\text{WinRate}(r_i) = \frac{1}{N-1} \sum_{j \neq i} \mathbf{1}[\text{Pref}(r_i, r_j) = 1] \quad (3)$$

and Spearman correlation is computed between win rates and human average scores.

5.2 RQ3 Results

Table 2 compares absolute scoring and pairwise preference learning against zero-shot LLM baselines. The pairwise model achieves $\rho = 0.688$, a 0.043 absolute Spearman gain over absolute scoring ($\rho = 0.645$). Because no significance test is performed on this difference, we do not claim statistical superiority. We interpret this as a consistent improvement from pairwise preference aggregation, though no significance test is performed on this difference. Among zero-shot models, Qwen2.5 72B ($\rho = 0.406$) and DeepSeek V3 ($\rho = 0.391$) show moderate positive correlation, while several larger or newer models perform poorly with the same prompt: Qwen3 235B ($\rho = 0.132$), DeepSeek V3.1 ($\rho = -0.049$), DeepSeek V3.2 ($\rho = -0.274$), GPT-4.1 ($\rho = -0.145$), and Kimi K2 ($\rho = -0.472$). These results suggest that task-specific supervision is more reliable than model scale alone for this evaluation problem.

Table 2: Cross-source pairwise preference aggregation achieves the highest ranking correlation in this benchmark. Results are reported without significance testing; pairwise evaluation requires $O(n^2)$ comparisons at inference.

Method	Params	Spearman ρ
Pairwise (cross source, ours)	8B	+0.688
Pairwise (cross source, ours)	7B	+0.665
Absolute scoring (balanced, ours)	8B	+0.645
Qwen2.5 72B Instruct (zero-shot)	72B	+0.406
DeepSeek V3 (zero-shot)	685B	+0.391
Qwen3 235B A22B (zero-shot)	235B	+0.132
DeepSeek V3.1 Terminus (zero-shot)	685B	-0.049
GPT 4.1 (zero-shot)	endpoint	-0.145
DeepSeek V3.2 (zero-shot)	685B	-0.274
Kimi K2 (zero-shot)	endpoint	-0.472

5.3 Zero Shot Model Scale and Ranking Correlation

Figure 5 plots parameter count against Spearman ρ for all pairwise zero-shot baselines. In our zero-shot prompting setup, larger model scale alone does not reliably improve rewriting evaluation. We do not interpret this as a general scaling law, since model families, training recipes, and prompting interfaces differ.

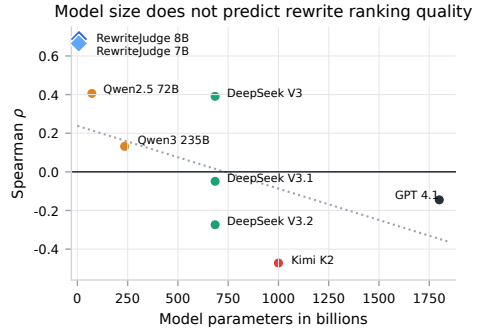


Figure 5: Zero-shot model scale and ranking correlation on RewritingBench. In our zero-shot prompting setup, larger model scale alone does not reliably improve correlation; we do not interpret this as a general scaling law because model families, training recipes, and prompting interfaces differ.

Within model families, newer or larger endpoints do not reliably outperform older ones in our zero-shot prompting setup (e.g., Qwen3 235B at $\rho=0.132$ vs. Qwen2.5 72B at $\rho=0.406$), though these systems differ in training and interfaces so this should not be read as a general generational claim (see Appendix E.1 for within-family comparisons).

Cross-source vs. same-source preference. Under the Qwen2.5 7B setting, the pairwise model trained on global cross-source comparisons reaches $\rho = 0.665$, while the

model trained on generated pairs from the same source reaches $\rho = 0.421$. The stronger cross-source result should be interpreted carefully. Cross-source comparisons pair rewrites from different source texts with different human scores and therefore cover a wider quality range, but they are less directly aligned with the practical setting of choosing among multiple rewrites for the same input. In our preliminary same-source setting, performance is lower, suggesting that same-source preference remains an open challenge rather than a solved problem. Pairwise evaluation for multiple rewrites of the same source remains an important setting for future data collection.

Why can pairwise help absolute scoring? We attribute this to two factors: (1) pairwise comparison may provide an easier local learning signal than direct score calibration, although cross-source comparison still requires a global notion of quality; (2) win rate aggregation provides a more robust quality estimate than any single absolute prediction, as each rewrite’s score is informed by multiple comparisons. This improvement comes at the cost of $O(n^2)$ comparisons at inference.

6 Robustness and Bias Analysis

The previous sections establish the core scientific claim: source-similarity metrics are negatively aligned with human rewrite quality, and task-specific supervision with comparative signals can help recover rankings that match human scores. We next summarize robustness checks and bias analysis.

6.1 Training Robustness

Training on the imbalanced score distribution causes mode collapse (Qwen3 8B: $\rho = 0.337$), while class-balanced oversampling (168 per score class) improves performance to $\rho = 0.645$, confirming that rewrite evaluation requires supervision across the full quality range. Rubric-aligned prompts improve absolute scoring ($\rho = 0.645$ vs. $\rho = 0.607$ standard), pairwise preference is numerically higher ($\rho = 0.688$), and reasoning prefixes hurt ($\rho = 0.360$). Additional ablations are reported in Appendix B.

6.2 Bias Checks: Length, Verbosity, and Position Effects

A reliable evaluator should judge quality based on content, not spurious features such as text length or verbosity. Table 3 compares bias across six methods along four dimensions. Both RewriteJudge variants show minimal output length bias (Qwen3 8B: $r = -0.033$; Qwen2.5 7B: $r = -0.058$), well below zero-shot Qwen3 8B ($r = 0.325$), character overlap ($r = 0.306$), and a length heuristic ($r = 0.287$). RewriteJudge Qwen3 8B achieves near-zero verbosity bias ($\rho = 0.006$), far below zero-shot Qwen3 8B ($\rho = 0.382$) and character overlap ($\rho = 0.760$).

Table 3: Bias analysis. Lower absolute correlation values indicate less bias. RewriteJudge shows low bias across all dimensions compared to baselines.

Method	Output Len. r	Input Len. r	Verbosity ρ	Position Bias r	Bias Score
RewriteJudge (Qwen3 8B)	-0.033	-0.070	0.006	0.120	0.057
RewriteJudge (Qwen2.5 7B)	-0.058	-0.115	0.106	0.187	0.116
Prometheus 2	0.010	-0.071	0.098	0.223	0.100
Zero-shot Qwen3 8B	0.325	0.167	0.382	0.437	0.328
Char Overlap	0.306	0.044	0.760	—	0.468
Length Heuristic	0.287	0.052	0.741	—	0.441

Position bias measures candidate-order preference in pairwise evaluation and applies only to methods that compare candidate pairs. RewriteJudge maintains low position bias (Qwen3 8B: $r = 0.120$; Qwen2.5 7B: $r = 0.187$), while zero-shot Qwen3 8B shows higher bias ($r = 0.437$). Char Overlap and Length Heuristic are non-pairwise methods where position bias is not defined. These results confirm that RewriteJudge’s ranking alignment with human scores is not driven by spurious length or verbosity features. Extended bias analysis with additional figures is provided in Appendix D.

As a preliminary downstream application, RewriteJudge can be used to filter generated rewrite candidates. Selecting the top 50% by model score yields higher model-assigned quality than random selection, while BLEU-based filtering performs close to random, consistent with the negative alignment found in Section 4. Because this analysis uses model-assigned scores rather than independent human ratings, full results are deferred to Appendix D.1.

7 Discussion and Limitations

Source similarity metrics reward outputs that stay close to the input wording; for rewriting, this is systematically misleading because a high-quality rewrite must both preserve meaning and transform expression, and these objectives can be inversely correlated. The negative correlations observed in RewritingBench are not random noise but follow from this fundamental mismatch in objectives. This implication extends beyond Chinese to any generation setting where good outputs should differ from the input while preserving intent.

RewriteJudge substantially improves ranking alignment over zero-shot and general-purpose evaluators, and bias analysis confirms it is not driven by length or verbosity features. However, this result demonstrates that task-specific supervision helps for this benchmark and task definition, not that RewriteJudge generalizes as a universal rewriting evaluator. Cross-source pairwise aggregation achieves the highest correlation ($\rho = 0.688$), but applies to global ranking rather than same-source selection and comes at $O(n^2)$ cost. Pairwise accuracy alone is also insufficient: Qwen3 235B reaches 72.0% pairwise accuracy yet only $\rho = 0.132$, while our fine-tuned 8B model achieves $\rho = 0.688$ (see Appendix E.1).

Scope and generalization. RewritingBench contains 730 Chinese general-purpose improvement pairs from a single-rewrite-per-source design; generalization to other languages, rewriting subtasks (style transfer, simplification), and same-source candidate selection remains unknown. The cross-source pairwise evaluation covers a wide quality range but requires a global notion of quality across different inputs and is not directly aligned with the practical setting of choosing among rewrites for the same source.

Validation and calibration. Our analysis focuses on ranking quality; better-calibrated numeric scores are an open direction. The filtering experiment (Appendix D.1) uses model-assigned scores rather than independent human ratings, so filtering gains should not be interpreted as human-validated quality improvements. Full-parameter tuning and larger annotated datasets may further improve performance beyond the LoRA adaptation used

here.

8 Conclusion

We introduced RewritingBench, a diagnostic benchmark for Chinese general-purpose rewrite quality evaluation. The benchmark shows that source-rewrite similarity can be negatively aligned with human judgments because high-quality rewriting requires both meaning preservation and appropriate expression transformation. Task-specific RewriteJudge training substantially improves ranking alignment compared with source-similarity metrics and zero-shot judges, and cross-source pairwise aggregation achieves the highest ranking correlation ($\rho = 0.688$). However, same-source preference evaluation, calibrated scoring, and human-validated filtering remain open directions. Overall, our results suggest that Chinese rewriting evaluation should measure transformation quality under semantic faithfulness rather than source similarity alone.

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A Training Configuration Details

Table 4 summarizes the training configuration for REWRITEJUDGE. We employ Low Rank Adaptation (LoRA) to fine-tune the model, injecting trainable matrices into seven target modules: query, key, value, output projection, and the three MLP layers (gate, up, and down projections). This configuration results in trainable parameters totaling approximately 0.2% of the full model.

For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA reparameterizes the update as:

$$W' = W_0 + \frac{\alpha}{r}BA \quad (4)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are trained while W_0 remains frozen.

A.1 Hyperparameters

The LoRA adapters are configured with rank $r = 16$, scaling factor $\alpha = 32$, and a dropout rate of 0.05. We fine-tune the model for 3 epochs using a learning rate of 2×10^{-4} with a cosine decay schedule (3% warmup). The training uses bf16 mixed precision. With a batch size of 4 and 4 gradient accumulation steps (effective batch size 16), the total training time is approximately 16 minutes on a single NVIDIA H20 GPU.

Table 4: Complete training configuration for RewriteJudge.

Hyperparameter	Value
Base model	Qwen3 8B, Qwen2.5 7B Instruct
Quantization	bf16 (no quantization)
LoRA rank (r)	16
LoRA scaling (α)	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	2×10^{-4}
LR scheduler	Cosine (warmup 3%)
Epochs	3
Batch size	4 (gradient accumulation: 4)
Effective batch size	16
Precision	bf16
Gradient checkpointing	Enabled
Training samples	1,008 (168 per class)
GPU	1× NVIDIA H20 GPU (96GB)
Training time	~16 minutes

B Pairwise Preference Ablations

B.1 LoRA Rank Ablation

Table 5 investigates the sensitivity of pairwise evaluation quality to LoRA rank (r). We use $r = 16$ as the pre-specified setting for all main experiments. The $r = 32$ results are reported only as a sensitivity analysis on the evaluation set and are not used for model selection. Performance is stable across ranks, with $r = 32$ achieving the highest Spearman ρ of 0.705 for Qwen2.5 7B and 0.713 for Qwen3 8B. The narrow range across ranks (maximal $\Delta\rho \approx 0.03$) indicates that pairwise preference learning is robust to adapter capacity.

Table 5: LoRA rank ablation: effect on trainable parameters and Spearman correlation.

Model	Rank r	α	Params	Spearman ρ
Qwen2.5 7B	8	16	~0.1%	+0.685
	16	32	~0.2%	+0.665
	32	64	~0.4%	+0.705
Qwen3 8B	8	16	~0.1%	+0.679
	16	32	~0.2%	+0.688
	32	64	~0.4%	+0.713

B.2 Pairwise Data Efficiency

Table 6 investigates the scaling behavior of pairwise evaluation quality. Qwen3 8B improves monotonically, reaching $\rho = 0.611$ with only 25% of the data and $\rho = 0.688$ with all 2,652 training pairs. Qwen2.5 7B shows an irregular result at 50% data caused by position bias, where the model frequently prefers the second presented rewrite regardless of content.

Table 6: Pairwise data efficiency: performance at different fractions of training data (LoRA $r = 16$, 3 epochs) for Qwen3 8B and Qwen2.5 7B.

Model	Training Data	Pairs	Spearman ρ
Qwen2.5 7B	25%	663	+0.434
	50%	1,326	+0.140
	100%	2,652	+0.665
Qwen3 8B	25%	663	+0.611
	50%	1,326	+0.669
	100%	2,652	+0.688

Position swap averaging mitigates this bias by presenting each pair in both orderings. A model that always predicts the second rewrite as better produces average preference ≈ 0.5 per pair, collapsing the win rate distribution and yielding near zero Spearman correlation. This makes it important to report both pair-

wise accuracy and ranking correlation for pairwise evaluators.

C Extended Results

C.1 Learning Curve with All Methods

Table 7 and Figure 6 show how evaluation quality changes as training data increases. Performance improves from 50 to 1,008 samples, with a clear jump after 400 samples. With 100 balanced samples, RewriteJudge reaches $\rho = 0.117$, already above all zero-shot baselines and general-purpose evaluators. The upward trend suggests that the model needs enough examples in each score class, approximately 168 samples here, to learn the distinctions required for strong rewriting evaluation.

Table 7: Learning curve comparison between Qwen2.5 7B and Qwen3 8B evaluators.

Samples	ρ (Qwen2.5 7B)	ρ (Qwen3 8B)
50	-0.026	+0.058
100	+0.094	+0.117
200	+0.259	+0.136
400	+0.235	+0.247
1008 (full)	+0.580	+0.645

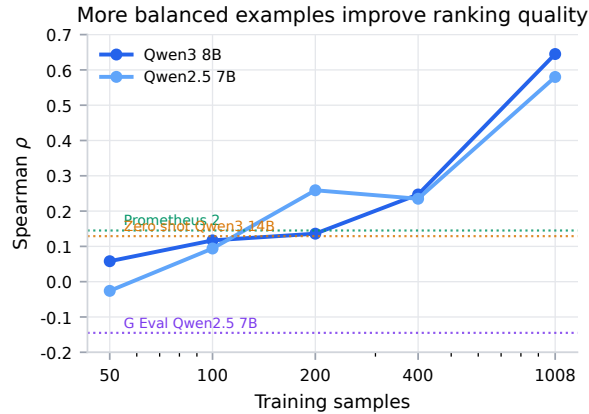


Figure 6: Full learning curve including both LoRA variants and baseline methods (dashed lines). All baselines show flat or negative correlation regardless of data size.

C.2 Score Distribution Analysis

Figure 7 shows the score frequency across different evaluators. Zero shot 7B has a ceiling effect: more than 80% of its predictions are Score 4. Prometheus 2 mostly predicts Score 3 and ignores much of the lower and upper score range. In contrast, REWRITEJUDGE, for both

Qwen3 8B and Qwen2.5 7B, produces a distribution closer to the human scores across all six score levels.

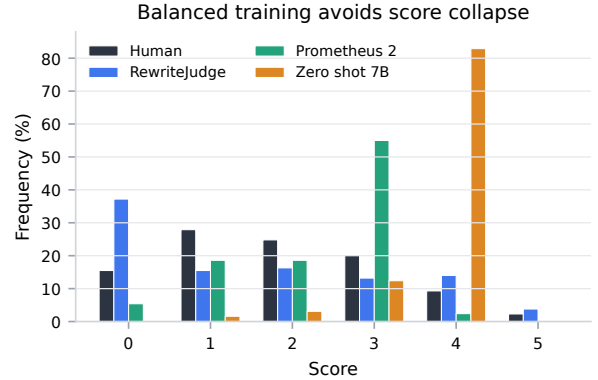


Figure 7: Distribution of human annotations and RewriteJudge predictions. Balanced training makes the model score distribution closer to the human score distribution.

C.3 Confusion Matrix Analysis

Figures 8, 9, and 10 provide a granular view of evaluator errors. Both REWRITEJUDGE variants exhibit a strong diagonal trend consistent with human judgment. The Qwen3 8B variant correctly identifies 60% of Score 0 samples and 66.7% of Score 5 samples, and shows a clear diagonal pattern from upper left to lower right with very few extreme misclassifications, such as predicting Score 5 for a human Score 0. The Qwen2.5 7B variant similarly concentrates predictions along the diagonal: 50% accuracy for Score 0 and 58.3% for Score 4, with the model showing particular strength in the middle and high score range. Both variants rarely commit extreme errors, confirming that class balanced fine tuning preserves ordinal structure.

In contrast, Prometheus 2 has very limited discriminative power: it predicts Score 3 for 63.6% to 73.9% of samples in every human score category. This mode collapse explains its low Spearman correlation ($\rho=0.145$), as it behaves almost like a constant score predictor.

D Extended Bias Analysis

This section provides additional bias analysis figures and details beyond the summary in Section 6.

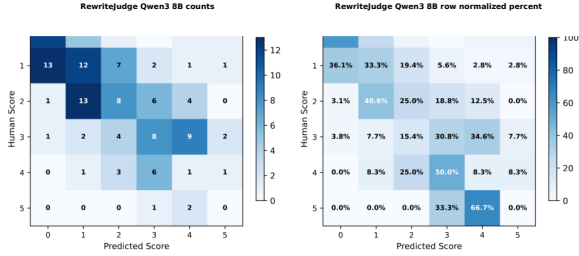


Figure 8: Confusion matrix for RewriteJudge(Qwen3 8B) predictions vs. human average scores (rounded). The model shows strongest agreement for extreme scores, including 0 and 4 to 5.

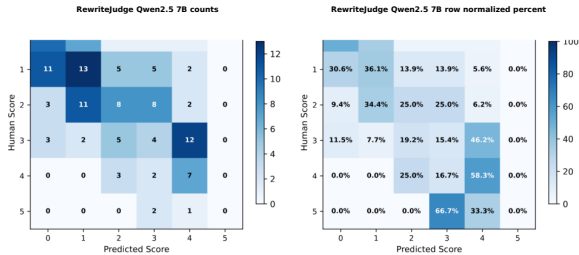


Figure 9: Confusion matrix for RewriteJudge (Qwen2.5 7B) predictions vs. human average scores (rounded). The model shows strongest agreement for extreme scores, including 0 and 4 to 5.

D.1 Preliminary Model-Scored Filtering Analysis

As a preliminary use case, we examine whether RewriteJudge can select candidates that receive higher RewriteJudge scores from a generated pool. This experiment is not an independent validation of human-perceived quality. We score 900 generated rewrites (300 sources $\times$ 3 quality levels) with RewriteJudge and then compare several selection rules in Table 8. The numbers in this table are assigned by RewriteJudge itself; they should be read as model scores, not as independent human ratings.

Selecting the top 50% by RewriteJudge score gives a mean model score of 4.25, while a random 50% subset has a mean model score of 2.90. The selected subset has higher model-assigned scores by construction, so this difference should not be interpreted as an independent quality improvement. BLEU-based filtering keeps 96.4% of the dataset and has a mean model score of 2.94, which is close to random selection, consistent with the earlier finding that overlap metrics are weak proxies

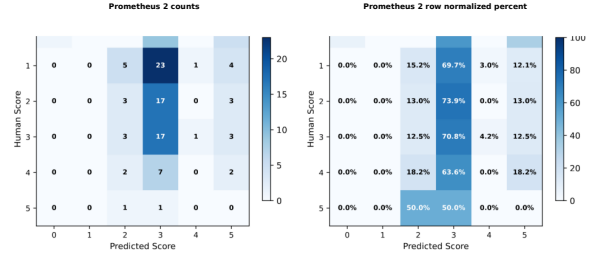


Figure 10: Confusion matrix for Prometheus 2. The model predicts Score 3 for the majority of samples across all human score categories, exhibiting mode collapse that explains its low Spearman correlation ($\rho = 0.145$).

Table 8: Preliminary filtering results scored by RewriteJudge itself. These scores are model outputs, not independent human ratings, and should not be interpreted as human-validated quality gains.

Strategy	N	Mean	σ	% ≥ 3
Top 30%	270	4.64	0.48	100.0
≥ 4 threshold	390	4.44	0.50	100.0
Top 50%	450	4.25	0.67	100.0
≥ 3 threshold	461	4.22	0.69	100.0
BLEU middle range	868	2.94	1.51	53.1
Random 50%	450	2.90	1.53	51.6
All (unfiltered)	900	2.87	1.54	51.2
Bottom 50%	450	1.48	0.67	2.4

for rewrite quality. We include two additional reference methods: (1) **character overlap** (JACCARD CHAR), representing traditional overlap metrics; and (2) a **length heuristic** that assigns quality scores based solely on the ratio of rewrite length to source length.

Length bias Figure 11 shows evaluator scores across output length quartiles: both RewriteJudge variants remain nearly flat, while zero-shot Qwen3 8B, character overlap, and the length heuristic all reward longer outputs regardless of quality. The length heuristic exhibits extreme verbosity bias ($\rho = 0.741$), confirming a known failure mode where naive metrics struggle to distinguish between informative expansion and redundant verbosity (Kuang et al., 2024).

Position bias Figure 12 summarizes all four bias dimensions as a compact heatmap: both RewriteJudge variants have low overall bias scores (0.057 and 0.116), while zero-shot Qwen3 8B (0.328), character overlap (0.468), and the length heuristic (0.441) show larger bias values across multiple dimensions.

Verbosity bias. Figure 13 breaks down

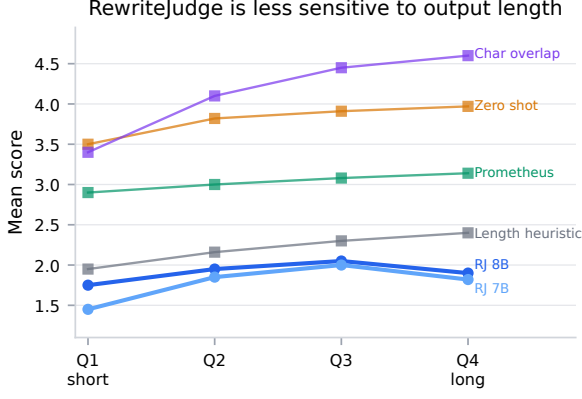


Figure 11: Mean evaluator score by output length quartile. RewriteJudge maintains consistent scores across quartiles, while traditional metrics show increasing scores for longer outputs.

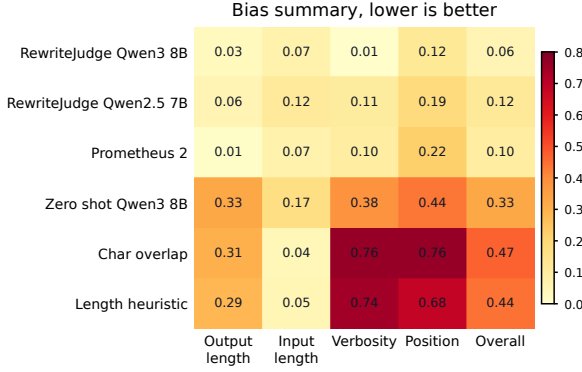


Figure 12: Bias heatmap across evaluator methods. Lower values indicate weaker correlation with spurious factors such as length, verbosity, or position.

verbosity bias by the ratio of output length to input length across two categories present in our evaluation set: Concise (ratio < 0.8) and Similar (0.8 to 1.2). RewriteJudge variants assign consistently low scores in both categories, remaining well below all other methods in absolute score level. In contrast, other methods show larger gaps: character overlap shows the largest jump ($3.4 \rightarrow 4.6$, $\Delta=1.2$), reflecting strong systematic preference for longer outputs. The verbosity bias scores in Table 3 (Section 6) confirm this pattern.

E Extended Zero Shot Baseline Analysis

E.1 Accuracy vs. Ranking Correlation

A natural expectation is that higher pairwise accuracy should correspond to higher Spearman ρ . Our results contradict this: Table 9 shows that pairwise accuracy and ranking cor-

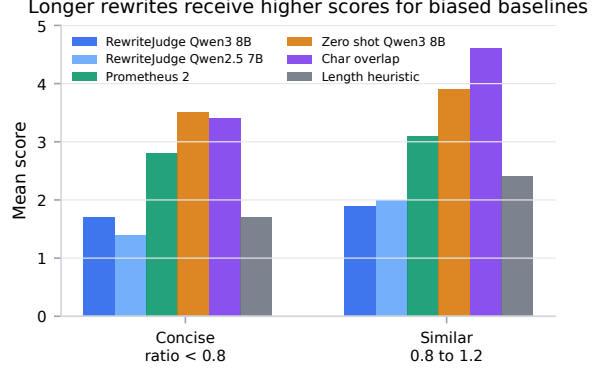


Figure 13: Verbosity bias comparison. RewriteJudge shows almost no verbosity bias, while the length heuristic exhibits extreme preference for verbose outputs ($\rho=0.741$).

relation are poorly aligned for zero-shot models.

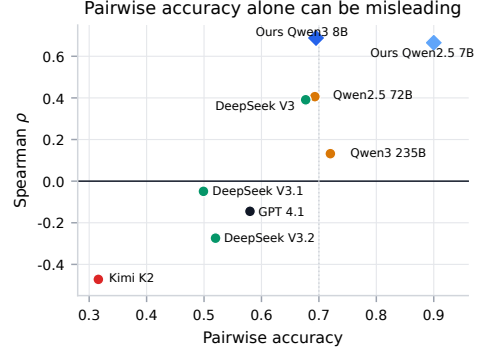


Figure 14: Pairwise accuracy vs. Spearman ρ : high accuracy but low ρ shows that accuracy can mislead for zero-shot evaluators.

As visualized in Figure 14, Qwen3 235B achieves high zero-shot accuracy (72.0%) yet fails to provide a meaningful ranking ($\rho = 0.132$). This suggests that zero-shot models may correctly identify easy pairs with large quality gaps but fail to maintain consistent logic across the full spectrum. Our fine-tuned model variants largely avoid this mismatch, as seen by their position in the upper right area of the figure.

E.2 Format Compliance vs. Evaluation Quality

We examine whether format compliance (the ability to follow the pairwise comparison output format) correlates with evaluation quality. Table 10 shows the parse failure rate for each zero-shot baseline.

The results in Figure 16 confirm that for-

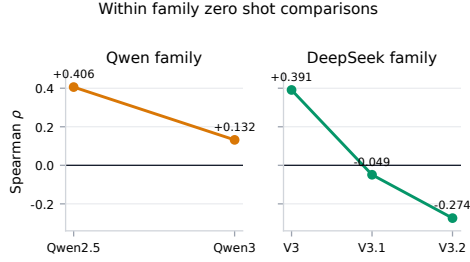


Figure 15: Within-family zero-shot comparisons. These results show that newer or larger endpoints are not reliably better in our prompting setup, but should not be read as a general generational trend.

Table 9: Mismatch between accuracy and correlation in zero-shot pairwise evaluation. Qwen3 235B achieves the highest accuracy (72.0%) but low $\rho=0.132$.

Model	Accuracy (%)	Spearman ρ	Paradox?
Ours (8B LoRA)	69.5	+0.688	No
Ours (7B LoRA)	90.0	+0.665	No
Qwen3 235B	72.0	+0.132	Yes
Qwen2.5 72B	69.3	+0.406	Mild
DeepSeek V3	67.7	+0.391	No
DeepSeek V3.1	49.9	-0.049	Yes
Kimi K2	31.6	-0.472	No

mat compliance and evaluation quality are **orthogonal**. Models with 100% compliance (0% failure) span the entire performance spectrum, from the best zero-shot baseline (DeepSeek V3) to the worst (Kimi K2). This indicates that the mechanical ability to follow instructions is independent of the critical capacity to judge text quality.

F Chinese Annotation Examples

To provide a qualitative understanding of the benchmark, Table 11 presents several representative cases covering various quality levels. These examples illustrate the nuances of Chinese text rewriting, ranging from simple lexical substitutions to sophisticated stylistic enhancements. We observe that REWRITE-JUDGE (RJ) predictions align closely with human gold standard scores, successfully capturing subtle improvements in fluency and expressiveness while remaining sensitive to instances where the rewrite fails to provide significant added value.

Table 10: Format compliance vs. evaluation quality. Models with zero parse failures include both the best and worst performers, indicating format compliance is independent of evaluation capability.

Model	Parse Fail (%)	Spearman ρ	Quality
DeepSeek V3	0.0	+0.391	Best zero-shot
Kimi K2	0.0	-0.472	Worst zero-shot
DeepSeek V3.1	0.0	-0.049	Poor
DeepSeek V3.2	0.0	-0.274	Poor
GPT 4.1	0.0	-0.145	Poor
Qwen2.5 72B	8.8	+0.406	Good
Qwen3 235B	9.0	+0.132	Poor

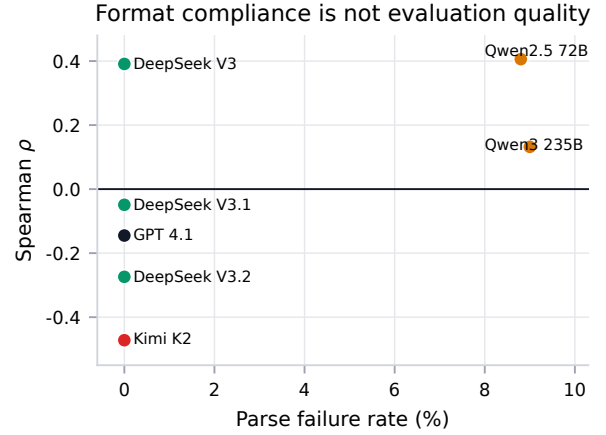


Figure 16: Parse failure rate vs. Spearman ρ . No correlation exists between format compliance and evaluation quality: models with 0% parse failures span the full quality range.

G Additional Error Analysis: Why Traditional Metrics Fail

Section 4 summarizes the core error mechanism in the main text. Here we provide additional detail on the two recurring patterns behind the negative correlation between overlap metrics and human judgment.

G.1 Two Error Patterns

We identify two systematic error patterns that explain the negative correlations:

Pattern 1: Surface similarity without quality. Rewrites that are close copies of the source text, changing only a few words while preserving the original structure, receive high overlap metric scores but low human ratings. For example, a rewrite that substitutes “加盟” with “加盟后” and “场场” with “几乎每场” achieves BLEU=0.55 but receives a human score of only 0.7. The metric rewards lexical overlap that reflects copying, not genuine rewrite quality.

Table 11: Representative examples from the evaluation benchmark. **Source**: original text; **Rewrite**: model generated rewrite; **Human**: average human score; **RJ**: RewriteJudge prediction.

Source	Rewrite	Human	RJ
今天天气很好，我们决定去公园散步。	阳光明媚，微风轻拂，我们来到公园漫步，享受这惬意的时光。	4	4
这个问题的解决方案需要进一步研究。	要解决此事还需深入探究才能找到答案。	2	2
他是一个非常好的学生，学习成绩优异。	他学习刻苦努力，是个品学兼优的好学生。	3	3
这家餐厅的菜很好吃。	这儿的菜品味道鲜美可口，令人回味无穷。	3	2
会议上讨论了关于环保的问题。	大家在会议中就环境保护议题展开了深入探讨。	4	4

Pattern 2: Quality without surface similarity. Strong rewrites that restructure the sentence, using different vocabulary and syntax while preserving meaning, receive very low metric scores but high human ratings. A rewrite that restructures a passage about a press conference into a more concise summary achieves BLEU=0.00 but receives a human score of 4.0. The metric penalizes the very lexical variation that characterizes successful rewriting.

G.2 Root Cause: Misaligned Objectives

These two patterns show that overlap metrics and human ratings answer different questions: overlap metrics measure how *similar* a rewrite is to its source, while human judgment evaluates how *good* the rewrite is. The critical insight is that these are not merely different metrics; they are *inversely correlated* for rewriting. Pattern 1 shows that high source similarity indicates *insufficient transformation* (the rewrite barely changed anything), while Pattern 2 shows that high source dissimilarity indicates *successful transformation* (the rewrite meaningfully transformed the text). This in-

verse relationship explains why the negative correlations are universal and statistically significant rather than reflecting random noise or a specific metric deficiency. For text rewriting, a task where the goal is to differ from the source while preserving meaning, these two objectives are inversely correlated. This explains why the strongest negative correlations are observed for character level metrics (JACCARD CHAR: $\rho=-0.60$), which most directly measure surface overlap, while semantic metrics like SBERT COSINE ($\rho=-0.28$) show somewhat weaker, but still negative, correlations.

G.3 Implications

This finding has practical implications for Chinese rewriting evaluation: source-rewrite similarity metrics are unreliable proxies for rewrite quality in our benchmark, and task-specific evaluators trained on human-rated data provide a more aligned alternative.

H Scoring Prompts for LLM and Human Evaluation

We investigate how prompt detail affects evaluation quality by comparing three variants: **Simple** (minimal), **Standard** (brief criteria), and **Detailed** (detailed rubrics). The **Detailed** prompt exactly matches our benchmark’s human rubric, grounding the model directly in the human annotation protocol.

Simple prompt

Simple prompt	
请帮我评判输入文本和输出文本重写的质量，并给出最终综合评分（0 到 5 分的整数）。	<i>Please evaluate the rewriting quality between the input and output texts, and provide a final overall score (integer from 0 to 5).</i>

Standard prompt

Standard prompt	
你是一个专业的中文文本改写质量评估专家。请根据以下维度对中文文本改写质量进行评分（0 到 5 分）： 评分维度： - 语义一致性：改写是否保留了原文的核心语义 - 句式重构：改写是否进行了句法结构的改变 - 词汇变化：改写是否使用了不同的词汇表达 - 风格保持：改写是否保持了原文的风格和长度 评分标准： 0 分：改写完全失败（严重语义扭曲或毫无改写） 1 分：改写质量很差 2 分：改写质量较差 3 分：改写质量一般 4 分：改写质量较好 5 分：改写质量优秀 请先简要分析，然后给出最终综合评分（0 到 5 分的整数）。	<i>You are a professional expert in evaluating Chinese text rewriting quality. Please score the rewriting quality from 0 to 5 along the following dimensions:</i> <i>Evaluation dimensions:</i> <i>Semantic consistency: Does the rewrite preserve the core meaning of the original?</i> <i>Syntactic reconstruction: Does the rewrite alter the syntactic structure?</i> <i>Lexical variation: Does the rewrite employ different vocabulary?</i> <i>Style preservation: Does the rewrite maintain the original style and length?</i> <i>Scoring criteria:</i> <i>0: Rewriting completely failed (severe semantic distortion or no change at all)</i> <i>1: Very poor rewriting quality</i> <i>2: Poor rewriting quality</i> <i>3: Moderate rewriting quality</i> <i>4: Good rewriting quality</i> <i>5: Excellent rewriting quality</i> <i>Please briefly analyze first, then provide a final overall score (integer from 0 to 5).</i>

Detailed prompt

Detailed prompt: identical to human annotation rubric	
请帮我判断输入文本与输出文本是否满足以下要求： • 要求 1. 在保持原文含义不变的前提下，用原文不同的文字进行表达，不额外新增、删减会严重影响语义的内容。 • 要求 2. 在保证逻辑正确、语序通顺的前提下，变换实体、短语、概念出现的前后顺序、结构关系；顺序变换幅度越小，分数越低。 • 要求 3. 在保证特殊实体、引用、解释、特定语句不变的前提下，使用同义词替换；同义词替换越少，分数越低。 • 要求 4. 保持与原文语言风格一致，字数长度变化合理；语言风格差距越大，分数越低。 • 要求 5. 改写综合评分要非常严格，同时考虑上述 4 个要求；在保证语义一致（要求 1）的前提下，（要求 2,3）变化越大越好，字数越接近越好（要求 4）；如果存在大量长字段完全复制，则给 0 分。 评分准则，每个要求最低 0 分，满分均为 5 分，评分要尽可能严格。	<i>Please evaluate whether the output text satisfies the following requirements with respect to the input text:</i> • Req. 1. <i>Express the same meaning as the original using different wording, without adding or omitting content that significantly affects semantics.</i> • Req. 2. <i>Rearrange the order and structural relations of entities, phrases, and concepts while ensuring logical correctness and fluency; the smaller the reordering, the lower the score.</i> • Req. 3. <i>Replace words with synonyms while keeping special entities, citations, explanations, and fixed expressions unchanged; fewer synonym substitutions yield a lower score.</i> • Req. 4. <i>Maintain a language style consistent with the original and a reasonable change in length; the greater the stylistic divergence, the lower the score.</i> • Req. 5. <i>Apply strict holistic scoring that simultaneously considers all four requirements above. Given semantic consistency (Req. 1), maximize structural and lexical changes (Reqs. 2 and 3) while keeping length close to the original (Req. 4). If large verbatim spans are copied, assign a score of 0.</i> <i>Scoring guideline: each requirement is scored from 0 (minimum) to 5 (maximum); scoring should be as strict as possible.</i>